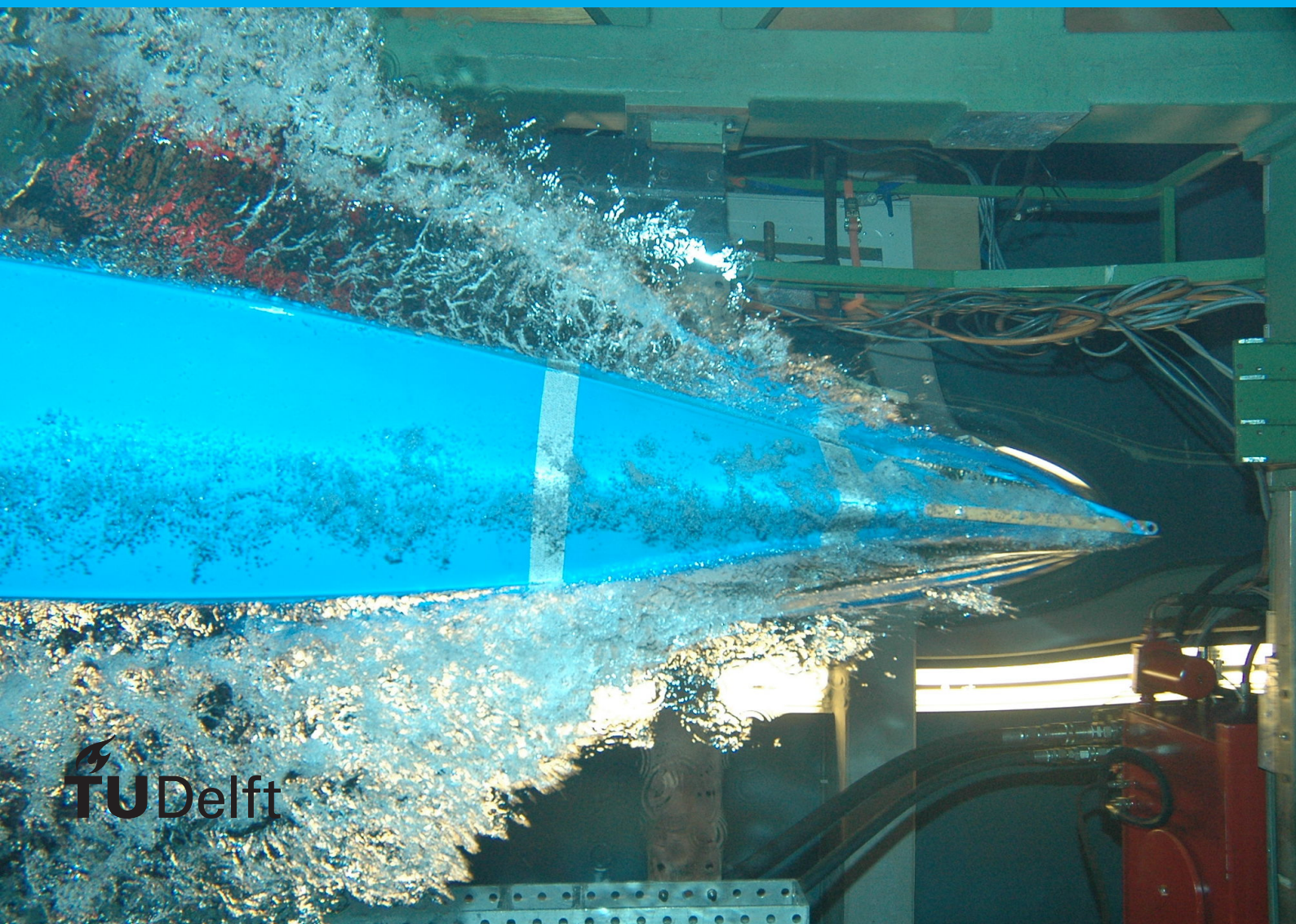
Title

Optional subtitle

J. Random Author

Cover Text
possibly
spanning multiple lines

ISBN 000-00-0000-000-0



Title

Optional subtitle

by

J. Random Author

to obtain the degree of Master of Science
at the Delft University of Technology,
to be defended publicly on Tuesday January 1, 2013 at 10:00 AM.

Student number:	1234567	
Project duration:	March 1, 2012 – January 1, 2013	
Thesis committee:	Prof. dr. ir. J. Doe,	TU Delft, supervisor
	Dr. E. L. Brown,	TU Delft
	Ir. A. Aaronson,	Acme Corporation

This thesis is confidential and cannot be made public until December 31, 2013.

An electronic version of this thesis is available at `http://repository.tudelft.nl/`.

TU Delft

Todo list

Rewrite this paragraph so it's easier to read. 1

Preface

Preface…

J. Random Author
Delft, January 2013

List of Figures

List of Tables

Acronyms

MDP Markov Decision Process.

RL Reinforcement Learning.

Contents

List of Figures vii

List of Tables ix

1 Introduction 1

2 Basics of Reinforcement learning 3

2.1 Finite Marcov Decision Process 3

3 Tabular Methods 5

4 Approximate solutions 7

5 Past, present and future of Reinforcement Learning 9

6 Proposed thesis research 11

7 Conclusion 13

Bibliography 15

1

Introduction

The technological advancements and steadily increasing demand for air and space travel of the past decades have lead to the development of new, more complex aerospace systems, resulting in new challenges in the field of aerospace control. At the same time advancements in automated control have been seen in many domains, ranging from industrial manufacturing and household appliances to the rapid widespread adoption of UAVs. However most of these systems are only able to complete their tasks within the narrow, predictable conditions they where designed for. Autonomous control systems need to be able to adapt to unexpected situations and environments. This has motivated researchers to search for new innovative methods that can be used to solve these new problems.

Rewrite this paragr it's easier to read.

Adaptive control methods allow autonomous systems to reconfigure them self in case of changing conditions. This opens the possibility to create fault tolerant control systems capable of safely recovering after a failure

One of the leading causes of fatal air traffic accidents is loss of control. If fully autonomous UAV's are to be used in the future, advanced flight control systems capable of adapting to rapidly changing conditions are necessary. This is because survival can only only be guaranteed if sta

2

Basics of Reinforcement learning

Reinforcement Learning (RL) is a method of machine learning different from other machine learning methods such as supervised learning which uses a set of labeled examples from a knowledgeable supervisor, and unsupervised learning which is typically used to find hidden patterns in unlabeled data. In Reinforcement Learning (RL) the learner is not told what action to take to achieve its goal, but instead it maps situations and actions by trying them out and discovering the most valuable. It does this by not only looking at the reward it received by it's latest action, but it also takes into account any future rewards it could receive in the future by taking that action.

2.1. Finite Marcov Decision Process

A Markov Decision Process (MDP) is a stochastic process that provides a mathematical framework used to describe problems where a learner learns from its actions to achieve an end goal. This framework is used in many areas, such robotics, economics, manufacturing and in the case of this report, Reinforcement Learning.

The two main components of an Markov Decision Process (MDP) are the agent, which makes decisions, interacts with its environment and learns. And the environment which consists of everything else. The two are constantly interacting with each other, the agent will perform an action which changes the state of the environment and the environment responds by giving it a reward. The goal of the agent is to maximize the reward it receives.

This happens in a sequence of discrete time steps. At the beginning of each time step the environment is at a certain state S. The agent then decides which action A to take which in turn changes the environment to the new state S'. In order for it to be a MDP this transition must satisfy the Marcov property, which means it needs to be memoryless and thus the environment's behavior must be independent of its past states. In other words, the probability of the environment to go from state S to S' and the rewards it gives back to the agent, must only be dependent on the initial state S and action A.

3

Tabular Methods

4

Approximate solutions

5

Past, present and future of Reinforcement Learning

6

Proposed thesis research

7

Conclusion

Bibliography

[1] G. J. Alred, C. T. Brusaw, and W. E. Oliu. *Handbook of Technical Writing*. St. Martin's, New York, 2003 (seventh edition).

[2] Michel Goossens, Frank Mittelbach, and Sebastian Rahtz. *The LaTeX Companion*. Addison-Wesley, Reading, Mass., 1997.

[3] S. F. Gull. Developments in maximum-entropy data analysis. In J. Skilling, editor, *Maximum Entropy and Bayesian Methods*, pages 53–71. Kluwer Academic, Dordrecht, 1989.

[4] K. M. Hanson. Introduction to Bayesian image analysis. In M.H̃. Loew, editor, *Medical Imaging: Image Processing*, volume 1898 of *Proc. SPIE*, pages 716–731, 1993.

[5] L. Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, Reading, Mass., 1994.

[6] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller. Equations of state calculations by fast computing machine. *J. Chem. Phys.*, 21:1087–1091, 1953.

[7] Frank Mittelbach, Michel Goossens, Johannes Braams, and David Carlisle. *The LaTeX Companion*. Addison-Wesley, Reading, Mass., 2004 (second edition).

[8] L. C. Perelman, J. Paradis, and E. Barrett. *Mayfield Handbook of Technical and Scientific Writing*. Mountain View, Mayfield, 1997. http://mit.imoat.net/handbook/.